AIM :

To write a Python program to solve the Water Jug Problem using the State Space Search technique.

ALGORITHM :

1. Start the program.
2. Initialize two water jugs with their capacities.
3. Set the initial state of both jugs as empty.
4. Define the goal state (required amount of water).
5. Generate possible operations:
    - Fill the first jug.
    - Fill the second jug.
    - Empty the first jug.
    - Empty the second jug.

- Pour water from one jug to another.

6. Apply the operations to reach the goal state.
7. Check whether the required amount of water is obtained.
8. Display the solution steps.
9. Stop the program.

PROGRAM :

pgrm3.py - C:/Users/nivet/OneDrive/Documents/pgrm3.py (3.11.0)

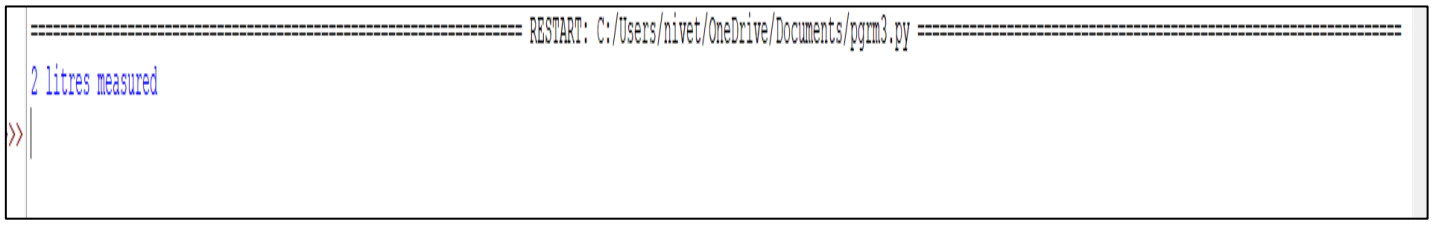
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```
a = 0
b = 0

while a != 2:
    if a == 0:
        a = 4
    elif b < 3:
        b = 3
    else:
        b = 0
        a = 2

print("2 litres measured")
```

OUTPUT :



RESULT :

Thus, the Python program to solve the Water Jug Problem using the State Space Search technique was executed successfully, and the required amount of water was obtained.